

EMCH 201: Introduction to Application of Numerical Methods

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Assignment 5

Assignment due date: **11:59 PM, Monday Nov 12th, 2021**
Point Possible: 100 (20+40+40)

Submission requirements:

- Detailed problem-solving process needs to be provided for full points
 - Late submission will not be graded
- Given three data points: $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$, prove the expression for the coefficient b_2 in the quadratic interpolation is

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

(20 points)

- Given the data

x	1	2	3	4
$y = f(x)$	0.5	0.3	0.25	0.21

Using the Newton's divided-difference interpolating polynomials to obtain cubic interpolation of the data.

(40 points)

- Given the same data in Question 2:

x	1	2	3	4
$y = f(x)$	0.5	0.3	0.25	0.21

Using the third order Lagrange interpolating polynomial to evaluate the y value when $x = 2.5$

(40 points)